SIMATS ENGINEERING

# SIMATS Online Programming Lab

DS PROGRAMMING

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QUESTIONS 10 / 10 completed

Linear Search

## LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total comparisons made, search status (Found/Not Found), and time taken in milliseconds.
Sample Input:
Enter 1 for manual input, 2 for

PROGRAM EDITOR

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```c
#include<stdio.h>
int main()
{
  int a[10],i,n,key;

  printf("enter number of elements:");
  scanf("%d",&n);
  printf("enter elements:");
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  printf("enter element to search");
  scanf("%d",&key);

  for(i=0;i<n;i++)
  {
    if(a[i]==key)
    {
      printf("Element found at position %d",i+1);
      return 0;

    }
```

OUTPUT

```
enter number of elements:enter elements:enter element to searchElement found at position 3
```

INPUT

```
5
10 20 30 40 50
30
```

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QUESTIONS 10 / 10 completed

Binary Search

## BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search iterations with mid-point calculations, (4) show final result with comparison count, and (5) verify the array remains sorted post-search. Sample Input/Output:

PROGRAM EDITOR

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```c
#include<stdio.h>
int binary(int a[],int low,int high,int key)
{
  if(low>high)
    return -1;
  int mid=(low+high)/2;

  if(a[mid]==key)
    return mid;
  else if(key<a[mid])
    return binary(a,low,mid-1,key);
  else
    return binary(a,mid+1,high,key);
}
int main()
{
  int a[10],n,i,key,pos;
  printf("Enter number of elements:");
  scanf("%d",&n);

  printf("Enter sorted elements:");
```

OUTPUT

```
Enter number of elements:Enter sorted elements:element not found
```

INPUT

```
5
10 20 30 40 50
60
```

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QUESTIONS 10 / 10 completed

Binary Search- Iter

## BINARY SEARCH-ITERATIVE

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day >=30°C and last day <=25°C. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; n<=50.

Sample Input: 6

22 28 31 24 29 32

Sample Output (after sort):

22 24 28 29 31 32

First >=30: index 4 (31)

Last <=25: index 1 (24)

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>

int main()
{
  int a[10],n,i,key;
  int low=0,high,mid;
  printf("enter number of elements:");
  scanf("%d",&n);
  printf("enter sorted elemets:");
  for(i=0;i<n;i++)
    scanf("%d",&key);
  printf("Enter element to search:");
  scanf("%d",&key);

  high=n-1;

  while(low<=high)
  {
    mid=(low+high)/2;
    if(a[mid]==key)
    {
```

OUTPUT

```
enter number of elements:enter sorted elemets:Enter element to search:element
not found
```

INPUT

```
4
60 80 79 09
9
```

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QUESTIONS 10 / 10 completed

Linear Search -Sor

## LINEAR SEARCH - SORT

Read n book page counts into an array. Sort ascending using any method. Use linear search to find the first book having >= 300 pages and the last book having <= 150 pages. Display page counts and indices (0-based), or ""Not found"". Validate sorted order; n<=50.

Sample Input: 7
120 450 275 310 140 500 200

Sample Output (after sort): 120 140 200 275 310 450 500

PROGRAM EDITOR C Run Save

```c
#include<stdio.h>
int main()
{
  int a[10],n,i,j,temp,key;
  printf("enter number of elements:");
  scanf("%d",&n);
  printf("enter elements:");
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  for (i=0;i<n-1;i++)
    for(j=i+1;j<n;j++)
      if(a[i]>a[j])
      {
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;

      }
  printf("sorted array:");
  for(i=0;i<n;i++)
    printf("%d",a[i]);
```

OUTPUT

```
enter number of elements:enter elements:sorted array:1020304050/nEnter element
to search:Element found at position 3
```

INPUT

```
5
40 10 50 20 30
30
```

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QUESTIONS 10 / 10 completed

Binary Search - An

## BINARY SEARCH - ARRAY

Write a C program to perform Binary Search on a sorted array of 10 integers. The program should accept the array elements and a target value from the user. It must display the position of the element and the total number of comparisons made.

Sample Input:
Enter 10 sorted integers:
5 12 18 23 34 45 56 67 78 90
Enter element to search:
45
Sample Output:
Element 45 found at position 6

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>

int main(){
  int a[10],n,i,key;

  printf("enter number of elements:");
  scanf("%d",&n);

  printf("enter elements:\n");
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  printf("enter element to search:");
  scanf("%d",&key);

  for(i=0;i<n;i++){
     if(a[i]==key){
        printf("element found at position %d",i+1);
           return 0;
     }
  }
```

OUTPUT

```
enter number of elements:enter elements:
enter element to search:element found at position 3
```

INPUT

```
5
10 20 30 40 50
30
```

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QUESTIONS 10 / 10 completed

Linear Search -Ten

## LINEAR SEARCH - TEMPERATURE

"Read n daily temperatures into an array. Use Linear Search to find the last day whose temperature is less than 20°C. Display temperatures, index (0-based), and value. If not found, print ""Not found"". Validate n ≤ 50.
Sample Input: 7
18 25 21 19 30 17 24

Sample Output: 18 25 21 19 30 17 24
Last <20: index 5 (17)"

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>

int main(){
  int temp[10],n,i,value;

  printf("enter number of temparature readings:");
  scanf("%d",&n);

  printf("enter temparatures:\n");
  for(i=0;i<n;i++)
    scanf("%d",&temp[i]);

  printf("enter temparature to search:");
  scanf("%d",&value);

  for(i=0;i<n;i++){
    if(temp[i]==value){
      printf("temparature found at reading %d",i+1);
      return 0;
    }
  }
```

OUTPUT

```
enter number of temparature readings:enter temparatures:
enter temparature to search:temparature found at reading 3
```

INPUT

```
5
25 28 30 32 35
30
```

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QUESTIONS 10 / 10 completed

Linear Search -Pric

## LINEAR SEARCH - PRICES

"Read n product prices into an array. Sort ascending using any method. Use linear search to find the first price >= ₹500 and the last price <= ₹200. Display prices and indices (0-based), or ""Not found"". Validate sorted order; n<=50.

Sample Input: 7
150 650 220 90 500 780 180

Sample Output (after sort): 90 150 180 220 500 650 780
First >=500: index 4 (500)

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>

int main(){
  int price[10],n,i,value;

  printf("enter number of prices:");
  scanf("%d",&n);

  printf("enter prices:\n");
  for(i=0;i<n;i++)
    scanf("%d",&price[i]);

  printf("enter price to search:");
  scanf("%d",&value);

  for(i=0;i<n;i++){
    if(price[i]==value){
      printf("price found at position %d",i+1);
      return 0;
    }
  }
```

OUTPUT

```
enter number of prices:enter prices:
enter price to search:price found at position 4
```

INPUT

```
5
100 250 300 450 500
450
```

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QUESTIONS 10 / 10 completed

Bin. Search-Iterativ

## BIN. SEARCH-ITERATIVE-TEMP

"Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day >=30°C and last day <=25°C. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; n<=50.
Sample Input: 6
22 28 31 24 29 32
Sample Output (after sort):
22 24 28 29 31 32
First >=30: index 4 (31)
Last <=25: index 1 (24)"

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>

int main(){
  int temp[10],n,i,key;
  int low=0,high,mid;

  printf("enter number of temparatures:");
  scanf("%d",&n);

  printf("enter sorted temparatures:\n");
  for(i=0;i<n;i++)
    scanf("%d",&temp[i]);
  printf("enter temparature to search:");
    scanf("%d",&key);

  high=n-1;

  while(low<=high){
    mid=(low+high)/2;

    if(temp[mid]==key){
```

OUTPUT

```
enter number of temparatures:enter sorted temparatures:
enter temparature to search:temparature found at reading 4
```

INPUT

```
5
20 25 30 35 40
35
```